

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1468

Reynolds Hill 138 kV
April 29, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1468. The Interconnection Customer proposes a 81.8 MW solar photovoltaic facility interconnecting at the Reynolds Hill 138 kV substation in the Kentucky area. The request was received on April 29, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 557 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$27,781,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Frontier-Yuba City Sta A 138.0 kV line	Spatsum Lornex-McClellan Ar 138.0 kV	130.5	405	24.4
Kettle-Country Club 2 138.0 kV	Spatsum Lornex-Latah Junction 138.0 kV	129.9	154	8.4
Westin-Elgin-Yuba City Sta A 138.0 kV	McClellan Ar-Spatsum Lornex 138.0 kV line	129.3	418	24.4
Westin-Elgin-McClellan Ar 138.0 kV	Yuba City Sta A-Westin-Elgin 138.0 kV L	126.0	359	19.2
Temple Station 243-Westin-Elgin Substation 138.0 kV	McClellan Ar-Westin-Elgin 138.0 kV	123.3	300	15.3
McClellan Ar-Spatsum Lornex 138.0 kV	Yuba City Sta A Substation-Westin-Elgin 138.0 kV L	122.9	464	26.2
Harvest Power Orlando-Frontier 138.0 kV Line	Latah Jct-Spatsum Lornex 138.0 kV	121.9	304	22.1
S-Cortez 138.0 kV L	Summit Lake Wps-Edna 138.0 kV Line	116.2	197	28.0
Spatsum Lornex-Country Club 2 Substation 138.0 kV	Latah Jct-Spatsum Lornex 138.0 kV	115.4	179	8.4
Temple Station 243-McClellan Ar 138.0 kV line	McClellan Ar-Westin-Elgin 138.0 kV line	114.7	295	15.3
Silver Bay Power-Hotel Rd 138.0 kV	Lee County Solid Waste Energy-Cooper 345.0 kV	111.1	208	12.2
Delano-Frontier 138.0 kV L	McClellan Ar-Spatsum Lornex 138.0 kV	105.4	176	6.4
Boonville-N City 138.0 kV	Ntcq-Mocapueblo 7101-7104 138.0 kV	102.9	195	24.1
Guadalupe-Mistersky 138.0 kV	Wissota Substation-Berry Street 138.0 kV	102.3	305	8.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Guadalupe-Mortan 138.0 kV line	Wissota-Berry Street 138.0 kV Line	102.0	324	8.1
Mainwaring-Mortan 138.0 kV	Berry St-Wissota 138.0 kV	102.0	327	8.1
Alibates-Latah Jct 138.0 kV	Harvest Power Orlando-Frontier 138.0 kV	100.4	399	25.3
Spatsum Lornex-Latah Junction 138.0 kV Line	Harvest Power Orlando-Frontier 138.0 kV line	100.1	411	25.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Frontier-Yuba City Sta A 138.0 kV line	Rebuild line with higher-capacity conductor	\$10,335,000	19% (\$2,004,000)
Country Club 2-Kettle 138.0 kV	Rebuild line with higher-capacity conductor	\$10,721,000	31% (\$3,288,000)
Yuba City Sta A-Westin-Elgin 138.0 kV	Rebuild line with higher-capacity conductor	\$23,375,000	19% (\$4,532,000)
Mcclellan Ar-Westin-Elgin 138.0 kV	Rebuild line with higher-capacity conductor	\$6,639,000	19% (\$1,283,000)
Temple Station 243-Westin-Elgin Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$5,211,000	19% (\$998,000)
Mcclellan Ar-Spatsum Lornex 138.0 kV	Rebuild line with higher-capacity conductor	\$8,632,000	19% (\$1,669,000)
Harvest Power Orlando-Frontier 138.0 kV Line	Rebuild line with higher-capacity conductor	\$29,085,000	16% (\$4,659,000)
South-Cortez 138.0 kV	Rebuild line with higher-capacity conductor	\$3,620,000	13% (\$463,000)
Spatsum Lornex-Country Club 2 138.0 kV Line	Rebuild line with higher-capacity conductor	\$8,637,000	31% (\$2,648,000)
Mcclellan Ar-Temple Sta 243 138.0 kV	Rebuild line with higher-capacity conductor	\$5,946,000	19% (\$1,139,000)
Hotel Road-Silver Bay Power 138.0 kV	Rebuild line with higher-capacity conductor	\$19,652,000	5% (\$1,049,000)
Frontier-Delano 138.0 kV L	Reconductor line	\$2,932,000	28% (\$815,000)
Boonville-N City 138.0 kV	Reconductor line	\$7,448,000	16% (\$1,162,000)
Mistersky-Guadalupe 138.0 kV line	Reconductor line	\$5,381,000	5% (\$291,000)
Guadalupe-Mortan 138.0 kV line	Reconductor line	\$5,850,000	5% (\$317,000)
Mainwaring-Mortan 138.0 kV	Reconductor line	\$6,722,000	5% (\$364,000)
Latah Junction-Alibates 138.0 kV	Reconductor line	\$3,985,000	15% (\$592,000)
Latah Junction-Spatsumlornex 138.0 kV	Reconductor line	\$3,419,000	15% (\$508,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$27,781,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.